



Patuakhali Science and Technology University

Faculty Of Computer Science and Engineering

CCE 310: Software Development Project-I

Project title – Cholo

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Submitted by	Submitted to
Tanmoy Kumar Das ID – 2102079 Reg – 10206	Prof. Dr. Md Samsuzzaman Department of Computer and Communication Engineering Arpita Howlader Department of Computer and Communication Engineering

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Cholo – Student Ride Sharing App

1. Abstract

In modern university campuses, many students commute daily using personal vehicles like bikes and cars. However, a significant number of these vehicles travel with empty seats, leading to wasted resources and increased traffic. The app Cholo is developed to address this issue by creating a student-only ride-sharing platform that makes campus commuting smarter, cheaper, and more efficient.

2. Objectives

The objective of this project is to design and develop a ride-sharing application tailored exclusively for university students. The app will:

- Allow students with vehicles to offer available seats for rides.
- Enable students without transportation to search and book rides on desired routes.
- Reduce commuting cost for students.
- Promote eco-friendly commuting by maximizing vehicle occupancy.

3. Problem Statement

Most students in universities commute using their own vehicles, often leaving one or more seats unused. Meanwhile, other students without transportation options rely on expensive services like Uber or Pathao. There is no existing solution that efficiently connects students to share rides within the campus environment, resulting in higher transportation costs and unnecessary traffic congestion.

4. Related Work

While there are several ride-sharing platforms like **Uber**, **Pathao**, and **Lyft**, none are designed to the specific needs of university campuses. Moreover, these platforms do not verify student identity or offer reduced pricing models designed for students. Some universities have experimented with internal ride boards or social media groups, but they lack real-time availability, route tracking, or booking mechanisms. Cholo bridges this gap with a real-time, verified, student-only ride-sharing system.

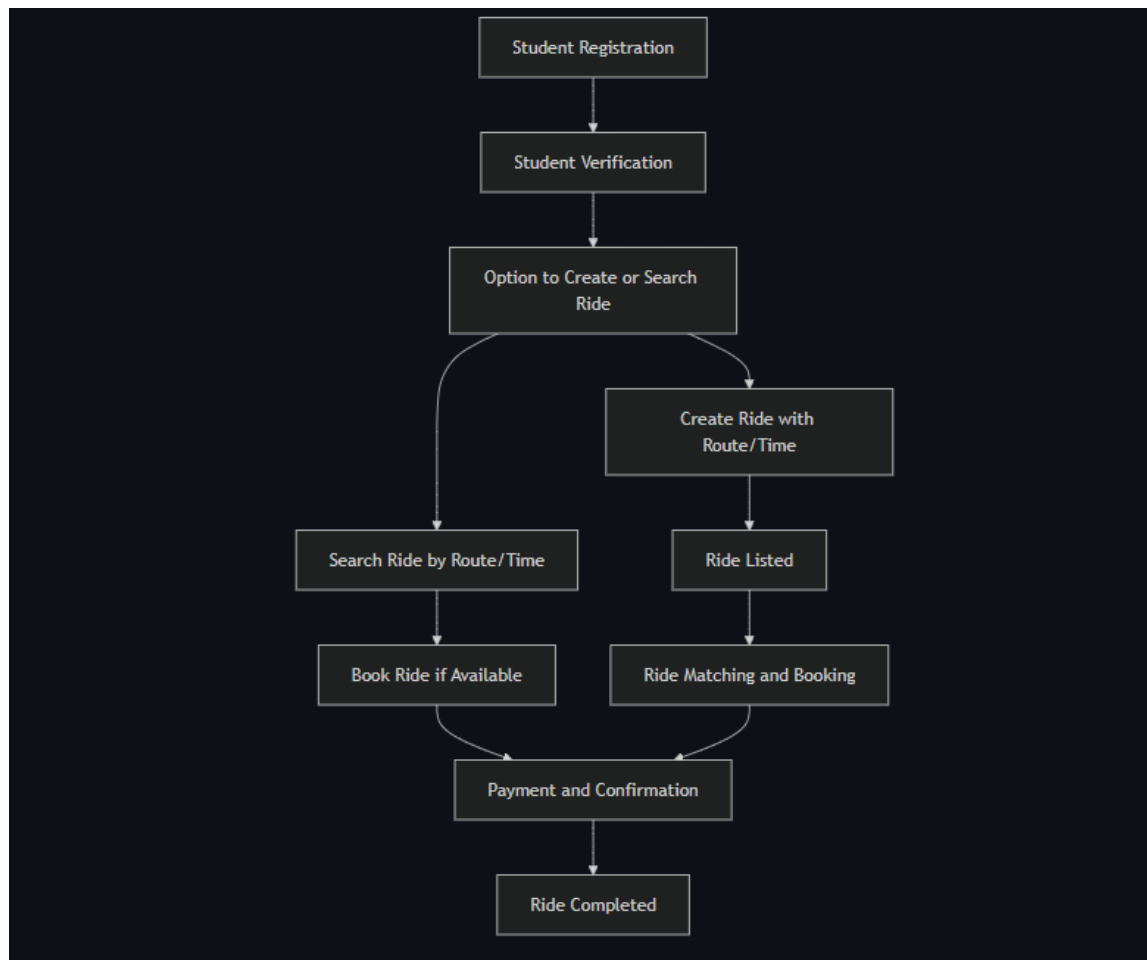
5. Methodology

Cholo is a mobile application built using Flutter, integrated with a backend using MongoDB, and developed in Android Studio.

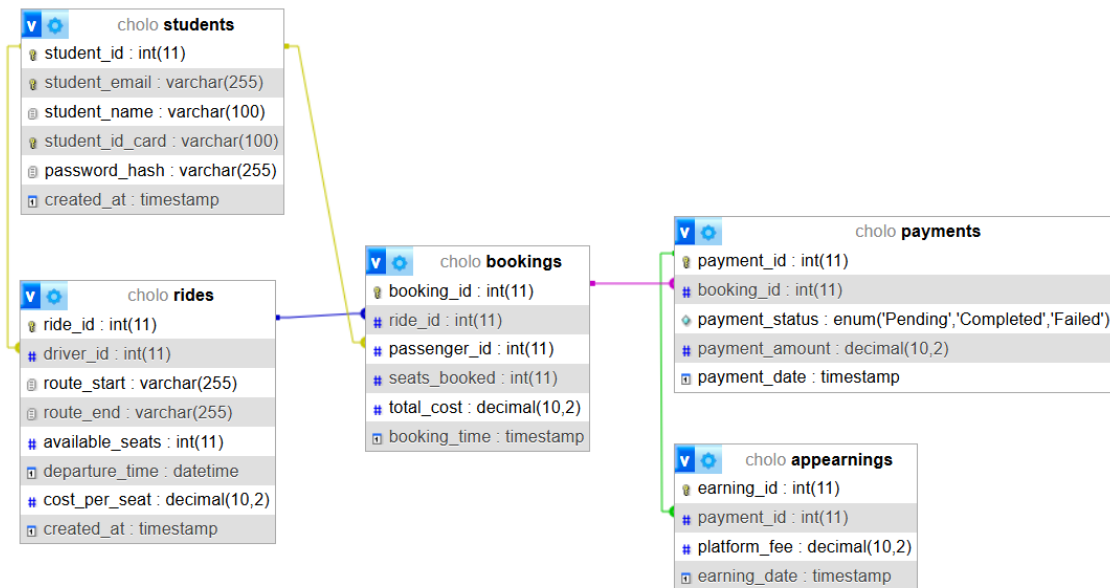
- User Verification: Only students can register, requiring student ID verification.
- Ride Offer: A student with a vehicle can list a ride with available seats, route, and time.
- Ride Booking: Another student can search available rides based on route and time, then book a seat if available.
- Payment Structure: Each ride is priced 40% less than Uber or Pathao, where 50% goes to the driver, 40% saves the rider, and 10% is platform fee for developers.
- Security Measures: Account verification and in-app messaging for secure communication.

6. Flow Chart

Below is a flow chart depicting the user journey in the Cholo app:



7. ERD (Entity Relation Diagram)



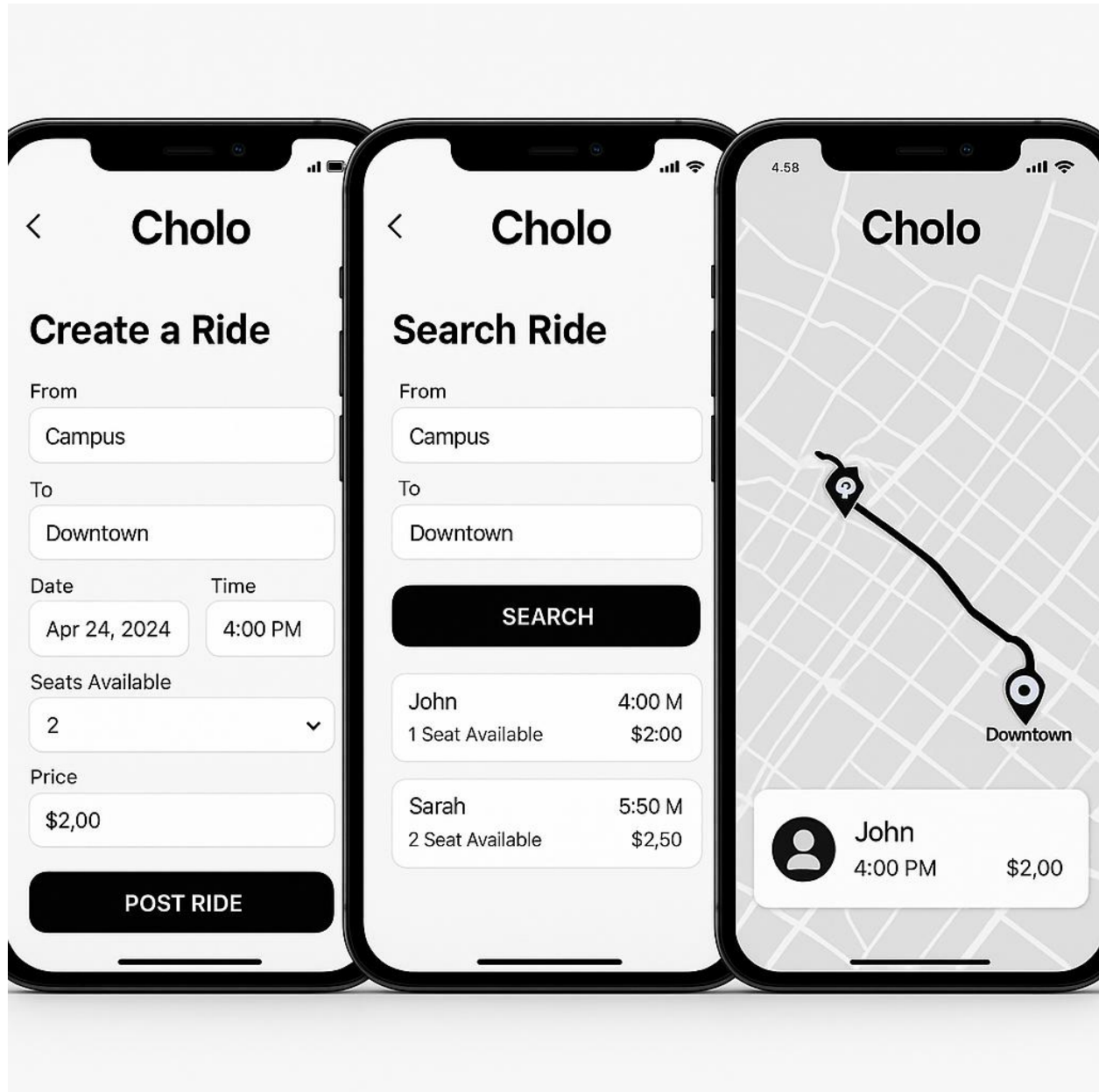
8. Gantt Chart

Below is the project timeline:

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Requirement Analysis	✓					
Designing UI/UX		✓				
Backend Development			✓	✓		
Integration				✓	✓	
Testing					✓	
Deployment						✓

9. UI (User Interface)

This is the initial user interface (UI) design. I am currently working on it, and it might look different in its final state. However, this represents the initial stage of development that I am focusing on.



10. Table of Work Progress

Mid-Term Progress

Task	Status	Notes
Basic UI Design	In Progress	Flutter UI designed for both roles
Student Login & Verification	In Progress	ID upload and verification logic
Ride Creation Form	In Progress	Route and seat input being tested

Final-Term Progress

Task	Status	Notes
Booking System Integration	In Progress	Seat booking with database update
Payment Flow Design	In Progress	Dummy payment system integrated
Ride Matching Algorithm	In Progress	Based on time, route
Deployment	In Progress	Final tests before release

11. Result and Limitations

Results:

- Successfully tested student registration and ride creation.
- Booking system shows working results with route-based filtering.
- UI is responsive and user-friendly.

Limitations:

- Real-time location tracking is not yet implemented.
- No in-app payment integration with bKash/Nagad (currently using dummy system).
- Requires active student community for scaling.

12. Conclusion

Cholo provides a practical, cost-effective, and eco-friendly solution for university students by connecting ride-givers and ride-seekers. It benefits both students saving cost for the passenger and earning for the driver. With future improvements like live GPS, in-app payments, and cross-campus partnerships, this app can evolve into a robust ride-sharing platform across multiple institutions.